

Adam Sayyed

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EDUCATION

University of Maryland, Baltimore County (UMBC) - Meyerhoff Scholar

June 2023 - Present

B.S. Computer Science, ML / AI Specialization, GPA: **3.8**

TECHNICAL SKILLS

Languages: Python, C++, Java, JavaScript, SQL, Bash, C#, HTML/CSS, R, Assembly, Haskell, C

Libraries/Frameworks: PyTorch, Kubeflow, Scikit-learn, Pydantic, Pandas, NumPy, SciPy, React, Polars, NodeJS

Developer Tools: Kubernetes, Docker, Git, GitLab, CI/CD, AWS, GCP, Azure, CUDA, SLURM, Grafana, OpenMP

EXPERIENCE

Johns Hopkins Applied Physics Laboratory, Software Engineering Intern

June 2026 - Present

- Built a FastAPI service used by 20 analysts to convert simulated sensor data into tactical formats, applying hundreds of schema validations and data transformations across hundreds of datasets and cutting manual data preparation by 65%.
- Enabled specialized tactical analysis by producing validated, analysis-ready datasets consumed by downstream metrics and analysis functions, allowing analysts to evaluate sensor data against operational requirements.
- Accelerated passive radar processing by 60% and reduced data-quality failures by 70% through scalable Python pipelines for data ingestion, validation, transformation, anomaly detection, and monitoring.

Sandia National Laboratories, Software Engineering & Machine Learning Intern

May 2025 - June 2026

- Developed and prototyped ML models (SVM, Random Forest, Gradient Boosting) in Python and C++ for geospatial data, boosting classification accuracy to 95% and enabling autonomous multi-object tracking for field operations
- Developed an IoU-based error-analysis pipeline that classified detections into TP/FP/FN, enabling systematic diagnosis of localization and classification failures across thermal datasets
- Architected Model Context Protocol (MCP) integrations with LLMs, exposing automated RESTful function calls that eliminated manual query workflows and accelerated end-to-end geospatial data analysis

University of Maryland Baltimore County, Undergraduate Researcher

January 2025 - Present

- Achieved 70% stock-movement prediction accuracy by building a multimodal ML pipeline integrating FinBERT sentiment embeddings with time-series indicators across 3 live data sources (Yahoo Finance, NewsAPI, GDELT)
- Eliminated data inconsistency across 50,000 stock trading day entries by implementing temporal alignment and preprocessing modules, ensuring clean feature handoff between ingestion and model training
- Fine-tuned FinBERT and RoBERTa models on finance-specific corpora to generate domain-adapted sentiment vectors, improving embedding quality for downstream stock-movement classification

PROJECTS

UMBC DoorDash Platform | React, Node.js, Express, MongoDB, WebSockets, REST APIs

- Engineered a full-stack campus food delivery system with React frontend, Node.js backend, MongoDB persistence, and real-time WebSocket notifications, increasing order throughput by 24% and enabling scalable transaction handling
- Implemented optimized scheduling, delivery-matching algorithms, and transaction logging, identifying peak-demand bottlenecks, optimizing worker utilization, and producing actionable operational insights for future scaling

High-Performance 3D Heat Diffusion Engine | C++, MPI, OpenMP, CUDA, Slurm

- Developed a 3D heat diffusion simulation with CPU, multi-node, and GPU execution, implementing MPI/OpenMP hybrid parallelism and achieving near-linear scaling with about 8 times speedup on 2 nodes for a 400³ grid
- Implemented CUDA acceleration and performed profiling of memory, cache, and communication bottlenecks, achieving about 1.6 times speedup on GPU versus CPU while identifying PCIe transfer limits

AWARDS

- Meyerhoff Scholar, UMBC
- University of Maryland Baltimore County Dean's List / President's List

January 2023 - Present

CONFERENCE PRESENTATIONS

- 2025 Joint Mathematics Meetings (JMM 2025)
- IOWA Summer Undergraduate Research Conference (SURC)

January 2025

July 2024